

YASIR ARFAT

Aspiring Software Engineer

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Summary

Computer Science undergraduate (2027) skilled in Full Stack Development (MERN), Machine Learning, and Generative AI. Experienced in building React/Node.js web platforms, FastAPI-based ML systems with explainable predictions, and RAG applications using AWS Bedrock and LangChain.

Education

- Malla Reddy Engineering College** **2023 – May 2027 (Expected)**
B.Tech in Computer Science (CGPA: 8.5/10) *Hyderabad, India*
- Kendriya Vidyalaya** **2021 – 2023**
Senior Secondary (77.5%) *Hyderabad, India*

Technical Skills

- Languages:** C++, JavaScript, Python, Java, SQL
- Frontend:** HTML, CSS, React.js
- Backend:** Node.js, Express.js, FastAPI, REST APIs
- Databases:** MongoDB, MySQL
- Machine Learning:** NumPy, Pandas, Scikit-learn, Matplotlib, Explainable AI
- Generative AI:** RAG, LangChain, FAISS, LLMs, AWS Bedrock
- Authentication & Security:** JWT, bcrypt, Role-Based Access Control (RBAC)
- Tools & Cloud:** Git, GitHub, Docker, AWS EC2, Postman, VS Code
- CS Fundamentals:** DSA, DBMS, OS, OOPS, Computer Networks

Projects

- Conversational RAG PDF Q&A System using AWS Bedrock** (Python, LangChain) [GitHub] [Live]
 - Designed and deployed a scalable conversational **RAG system** enabling semantic search across multi-document PDFs.
 - Implemented **FAISS-based vector indexing** with **Amazon Titan embeddings** for efficient similarity retrieval.
 - Integrated **LLaMA-3 via AWS Bedrock** to generate context-aware responses with reduced latency.
 - Deployed on **AWS EC2** with secure **IAM** configuration ensuring production-ready access control.

Student Marks Prediction System (Python, Scikit-learn, FastAPI, MySQL, React) [GitHub]

- Built an end-to-end ML system using **15,000 student records**, applying NumPy, Pandas, and Scikit-learn for data preprocessing, feature engineering, and model training.
- Trained and compared **four regression models: Linear Regression, Decision Tree, Random Forest, and Gradient Boosting**, using cross-validation and hyperparameter tuning to achieve an **R² score of 0.857** with the final model.
- Developed a **FastAPI + MySQL** backend using SQLAlchemy, JWT authentication, bcrypt, and RBAC, with a React dashboard supporting **Explainable AI, What-If analysis, prediction history, personalized recommendations, and admin analytics**.
- AI-Powered Interview Preparation Platform** (React, Express.js, Socket.io, Gemini AI) [GitHub] [Live]
 - Built an AI-powered interview preparation platform for HR, Technical, and Project-based practice.
 - Implemented resume upload with **ATS score generation** and **semantic search** for efficient resume analysis.
 - Developed real-time mock interviews using **Socket.io** and **Gemini AI** for dynamic conversational interactions.
 - Created a centralized placement resources hub with aptitude, interview, and learning materials.

Achievements

- IIT Hyderabad:** Contributed to the Campus-Connect project using MERN Stack and Google Gemini AI.
- MRDU Hackathon:** Selected among the Top 10 teams out of 500+ teams for fine-tuning an AI image model.